Supplementary Material for "A pre-registered test of cross-fitted stacking with a novelty-damped local residual correction on the Matbench composition tasks"

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S1. Pre-registration and amendment

The analysis plan (`ANALYSIS_PLAN.md`) and SHA-256 hashes of all code were committed before any official test fold was scored.

- **Original freeze:** `results/freeze_hashes_original_plan.txt`. Under it, the design was fixed on inner CV of fold 0's training rows.
- **Amendment 1:** `results/freeze_hashes.txt`, plan Section 8. Written before any scoring. It moved the choice among the explored design configurations inside every official training fold, because fold 0's training rows contain the test rows of folds 1–4.
- **Superseded run:** a first run under the original plan produced test predictions that were never scored. They are archived unscored in the repository.
- **Deviations:** all are listed in plan Section 7. They are a one-line fix in the scoring script (iterating over each fold's own arms), scheduling-only changes, and reporting scripts.

S1b. Amendment 2

Amendment 2 (plan Sections 9–11, `results/freeze_hashes_amendment2.txt`) was written after the Amendment-1 test results had been seen. It changed the expert pool, the features (including a disclosed DFT transfer block) and the correction gate. It was frozen before any Amendment-2 test prediction, and the test folds were scored once under it. The design evidence (fold 0 training rows only) is in `results/design_a2/DESIGN_LOG_A2.md`. The leaders' per-sample predictions used for the paired comparison come from the public `matbench` repository (`benchmarks/matbench_v0.1_{TPOT,darwin,modnet_v0.1.12}/r`

S2. Per-fold design selection (Amendment 1)

Table 1 lists the candidate selected inside every official training fold, with the cross-fitted OOF score of the uncorrected global stack for each candidate (MAE for regression; ROC-AUC for classification). Regression candidates:

- A: base features, L2 stack;
- B: base features, L1 stack, plus an L1-objective LightGBM expert;
- D: extra features, L1 stack;
- C: extra features, L1 stack, plus the L1-LightGBM expert.

Classification candidates: A (base features) and C (extra features).

Table 1: Per-fold design selection (inside each official training fold).

Task	Fold	Selected design	Candidate stack OOF scores
steels	0	B_base_l1	A=84.3708, B=83.1777, D=85.4401, C=85.4401
steels	1	D_extra_l1	A=90.1208, B=89.8331, D=89.2622, C=89.2622
steels	2	A_base_l2	A=92.3679, B=94.5285, D=93.8040, C=93.8040
steels	3	C_extra_l1_lgbml1	A=92.6136, B=92.0424, D=91.4351, C=91.4351
steels	4	B_base_l1	A=88.8951, B=88.8950, D=89.2953, C=89.2953
expt_gap	0	D_extra_l1	A=0.4159, B=0.4121, D=0.3408, C=0.3408
expt_gap	1	D_extra_l1	A=0.4192, B=0.4180, D=0.3408, C=0.3409
expt_gap	2	D_extra_l1	A=0.4067, B=0.4021, D=0.3411, C=0.3411
expt_gap	3	D_extra_l1	A=0.4175, B=0.4185, D=0.3451, C=0.3454
expt_gap	4	D_extra_l1	A=0.4115, B=0.4062, D=0.3385, C=0.3385
expt_is_metal	0	C_extra	A=0.9739, C=0.9819
expt_is_metal	1	C_extra	A=0.9733, C=0.9808
expt_is_metal	2	C_extra	A=0.9746, C=0.9813
expt_is_metal	3	C_extra	A=0.9725, C=0.9795
expt_is_metal	4	C_extra	A=0.9743, C=0.9812
glass	0	C_extra	A=0.9458, C=0.9461
glass	1	A_base	A=0.9521, C=0.9520
glass	2	C_extra	A=0.9515, C=0.9523
glass	3	C_extra	A=0.9508, C=0.9508
glass	4	C_extra	A=0.9449, C=0.9453

S3. Ideation and prior-art process

Literature search, prior-art checking and idea generation used the ResearchStudio Idea skills (paper-search, scoop-check, idea-spark), driven by an AI assistant; all records are in the repository (`research/`, `ideaspark_run/`).

- **Sources.** Retrieval used arXiv, OpenAlex, Semantic Scholar and Crossref. DBLP failed to return parseable responses, and OpenReview returned no records (rate-limited and off-venue).
- **Prior-art verdict: medium overlap.**
  - No retrieved work combines cross-fitted heterogeneous stacking, a novelty-damped local OOF correction and the Matbench composition tasks.
  - Every component has ancestors: stacking, regression kriging and Δ-learning.
  - The closest external works are a stack with a global kernel-ridge residual correction in operator learning, MODNet, and a stacked band-gap ensemble.
- **Idea card.** idea-spark proposed the fold-matched correction. Its coherence check verified the decomposition naive = analogue + absorption on synthetic instances, and its independent post-revision review passed.
- **Run outcome.** The run ended in a `needs_work` state because of an internal inconsistency in the tool's novelty re-audit gate. We therefore treated the proposal as unvalidated and pre-registered it as a secondary arm.

S4. Design-phase log

All configurations explored on fold 0's training rows (five pseudo-folds) are listed in `results/design/DESIGN_LOG.md`. These numbers informed only which candidates exist, not the test-fold results.

S5. Leaderboard snapshot and label-based classification scores

The leaderboard values used (`results/leaderboard_snapshot.json`) were taken from the per-task Matbench pages on 18 October 2026.

For every listed classification entry, the reported mean "ROC-AUC" equals the mean balanced accuracy to four decimals. This is what `matbench` computes when boolean labels are recorded: ROC-AUC computed from hard labels equals balanced accuracy.

We provide two unsubmitted result files: one with labels at the inner-CV threshold (comparable with the board) and one with probabilities.